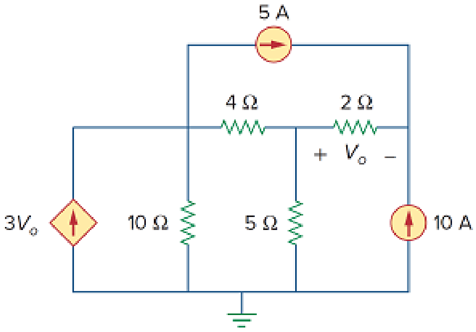
Problem

Obtain the node-voltage equations for the circuit in Fig. 3.111 by inspection. Then solve for *Vo*.

Figure 3.111:



Step-by-step solution

Step 1 of 7

Consider the following circuit diagram:

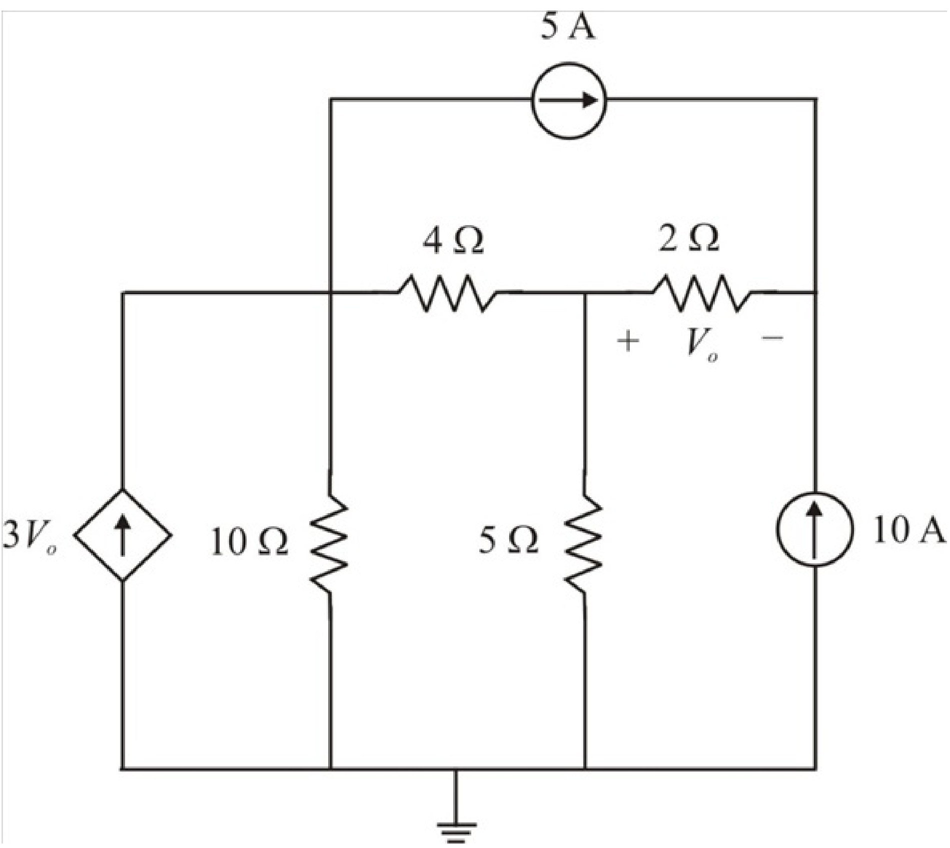


Figure 1

Step 2 of 7

Label the nodes on the circuit as shown in Figure 2:

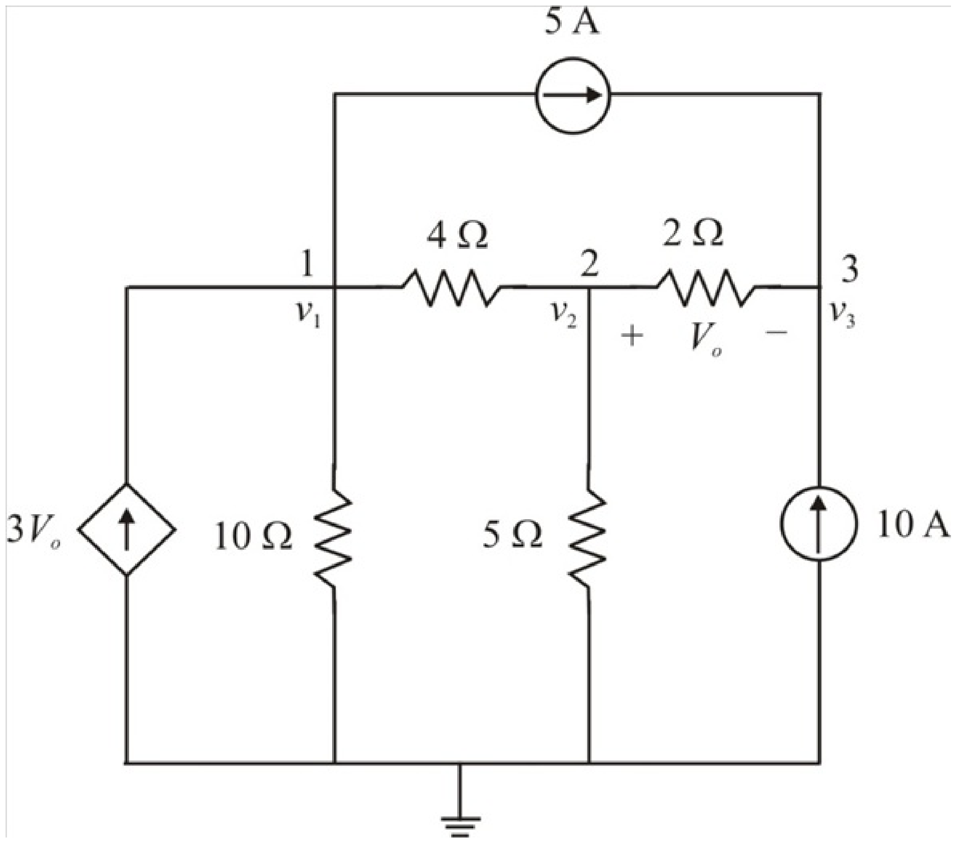


Figure 2

Step 3 of 7

If a circuit has *N* non-reference nodes, the node-voltage equations can be expressed in terms of conductance:

$$\begin{pmatrix} G_{11} & \cdots & G_{1N} \\ \vdots & \ddots & \vdots \\ G_{N1} & \cdots & G_{NN} \end{pmatrix} \begin{pmatrix} v_1 \\ \vdots \\ v_N \end{pmatrix} = \begin{pmatrix} i_1 \\ \vdots \\ i_N \end{pmatrix}$$ or simply G**v=i**.

Where,

Sum of the conductance connected to node *k* is G_{kk}

Negative of the sum of the conductance directly connecting nodes *k* and *j*, $k \neq j$ is $G_{kj} = G_{jk}$

Unknown voltage at node *k* is i_k

Sum of all independent current sources directly connected to node *k*, with currents entering the node treated as positive is v_k.

Step 4 of 7

Each of the diagonal terms is the sum of the conductance in the related to node, while each of the off-diagonal terms is the negative of the conductance common to nodes.

The circuit in Figure 1 has three non-reference nodes, so we need three node equations. This implies that the size of the conductance matrix **G**, is 3 by 3.

Find the diagonal terms of **G** in Siemens:

$$\begin{aligned}G_{11} &= \frac{1}{4}+\frac{1}{10}\\ &= 0.35\end{aligned}$$

$$\begin{aligned}G_{22} &= \frac{1}{4}+\frac{1}{2}+\frac{1}{5}\\ &= 0.95\end{aligned}$$

$$\begin{aligned}G_{33} &= \frac{1}{2}\\ &= 0.5\end{aligned}$$

Find the off-diagonal terms in the first row:

$$\begin{aligned}G_{12} &= -\frac{1}{4}\\ &= -0.25\end{aligned}$$

$$G_{13} = 0$$

Find the off-diagonal terms in the second row:

$$\begin{aligned}G_{21} &= -\frac{1}{4}\\ &= -0.25\end{aligned}$$

$$\begin{aligned}G_{23} &= -\frac{1}{2}\\ &= -0.5\end{aligned}$$

Find the off-diagonal terms in the third row:

$$G_{31} = 0$$

$$\begin{aligned}G_{32} &= -\frac{1}{2}\\ &= -0.5\end{aligned}$$

The input current vector **i** has the following terms, in amperes:

$$i_1 = 3v_o - 5$$

$$i_2 = 0$$

$$\begin{aligned}i_3 &= 10+5\\ &= 15\end{aligned}$$

Therefore, write the node-voltage equations:

$$\begin{pmatrix} 0.35 & -0.25 & 0 \\ -0.25 & 0.95 & -0.5 \\ 0 & -0.5 & 0.5 \end{pmatrix}\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 3v_o - 5 \\ 0 \\ 15 \end{pmatrix}$$

Write the first row equation:

$$\begin{aligned}0.35v_1 - 0.25v_2 &= 3v_o - 5\\ 0.35v_1 - 0.25v_2 &= 3(v_2 - v_3) - 5\\ 0.35v_1 - 0.25v_2 &= 3v_2 - 3v_3 - 5\\ 0.35v_1 - 3.25v_2 + 3v_3 &= -5\end{aligned}$$

Therefore, the node-voltage equations are $\boxed{\begin{pmatrix} 0.35 & -3.25 & 3 \\ -0.25 & 0.95 & -0.5 \\ 0 & -0.5 & 0.5 \end{pmatrix}\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} -5 \\ 0 \\ 15 \end{pmatrix}}$.

Use Cramer's rule to find the determinant Δ of the matrix:

$$\Delta = \begin{vmatrix} 0.35 & -3.25 & 3 \\ -0.25 & 0.95 & -0.5 \\ 0 & -0.5 & 0.5 \end{vmatrix}$$
$$= 0.35(0.475-0.25)+0.25(-1.625+1.5)+0$$
$$= 0.35(0.225)+0.25(-0.125)$$
$$= 0.0475$$

Use Cramer's rule to find determinant Δ_1 of the matrix:

$$\Delta_1 = \begin{vmatrix} -5 & -3.25 & 3 \\ 0 & 0.95 & -0.5 \\ 15 & -0.5 & 0.5 \end{vmatrix}$$
$$= -5(0.475-0.25)-0+15(1.625-2.85)$$
$$= -1.125-18.375$$
$$= -19.5$$

Step 6 of 7

Use Cramer's rule to find determinant Δ_2 of the matrix:

$$\Delta_2 = \begin{vmatrix} 0.35 & -5 & 3 \\ -0.25 & 0 & -0.5 \\ 0 & 15 & 0.5 \end{vmatrix}$$
$$= 0.35(0+7.5)+0.25(-2.5-45)+0$$
$$= 2.625-11.875$$
$$= -9.25$$

Use Cramer's rule to find determinant Δ_3 of the matrix:

$$\Delta_3 = \begin{vmatrix} 0.35 & -3.25 & -5 \\ -0.25 & 0.95 & 0 \\ 0 & -0.5 & 15 \end{vmatrix}$$
$$= 0.35(14.25+0)+0.25(-48.75-2.5)+0$$
$$= 4.9875-12.8125$$
$$= -7.825$$

Step 7 of 7

Calculate the node-voltage v_1:

$$\begin{aligned} v_1 &= \frac{\Delta_1}{\Delta} \\ &= \frac{-19.5}{0.0475} \\ &= -410.52\ \text{V} \end{aligned}$$

Calculate the node-voltage v_2:

$$\begin{aligned} v_2 &= \frac{\Delta_2}{\Delta} \\ &= \frac{-9.25}{0.0475} \\ &= -194.73\ \text{V} \end{aligned}$$

Calculate the node-voltage v_3:

$$\begin{aligned} v_3 &= \frac{\Delta_3}{\Delta} \\ &= \frac{-7.825}{0.0475} \\ &= -164.73\ \text{V} \end{aligned}$$

Calculate the voltage V_o:

$$\begin{aligned} V_o &= v_2 - v_3 \\ &= -194.73 - (-164.73) \\ &= -30\ \text{V} \end{aligned}$$

Therefore, the voltage V_o is $\boxed{-30\ \text{V}}$.